

A Hardware-Aware Deep-Learning based Navigation for On-Chip Retinal Prosthesis

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Abstract—Epiretinal prostheses restore partial vision to patients with photoreceptor degeneration, yet current systems, such as Argus II, rely on bulky waist-mounted vision processing units (VPUs) with no deep-learning-based scene understanding, fundamentally limiting navigation utility. This paper presents a hardware-aware co-design framework targeting a fully *spectacles-form-factor* VPU. A streaming end-to-end pipeline—sRGB-to-Y conversion, 640×480 to 60×80 spatial downsampling, gamma correction, denoising, and semantic segmentation—operates without any full-frame buffer. The segmentation engine, *NanoSeg*, is a 12,880-parameter CNN classifying pixels into four navigation-relevant classes (background, traversable, obstacle, hazard) and was developed through seven documented iterative architecture refinements on the SANPO pedestrian dataset. Under INT8 post-training quantization the total on-chip memory footprint estimate is 29.0 KB. FPGA C-synthesis on the Xilinx Zynq-7020 via Vitis HLS confirms a 67.2 ms inference latency at 100 MHz with 46% BRAM and 66% LUT utilization, demonstrating feasibility within wearable power and area budgets.

Index Terms—epiretinal prosthesis, semantic segmentation, streaming architecture, hardware-aware neural network, FPGA, post-training quantization, neural interface

I. INTRODUCTION

Retinal degenerative diseases such as retinitis pigmentosa (RP) and age-related macular degeneration (AMD) progressively destroy photoreceptor cells while leaving the inner retinal circuitry largely intact. AMD alone affects an estimated 196 million people worldwide, and is projected to reach 288 million by 2040 [1], while RP affects approximately 1.5 million individuals [2]. Together, these conditions represent the leading causes of irreversible vision loss in the developed world.

Epiretinal prostheses offer a promising intervention by stimulating residual retinal ganglion cells with spatially encoded electrical pulses, generating phosphene percepts that restore coarse functional vision. The Argus II system, the most widely deployed clinical device, demonstrated meaningful mobility improvements with a 60-electrode array [3], but its belt-worn vision processing unit (VPU) converts camera frames into simple brightness maps with no semantic scene understanding—it cannot distinguish a walkable surface from an obstacle or a traffic hazard. Second Sight’s cessation of operations in 2020 [4] further underscores the need for next-generation

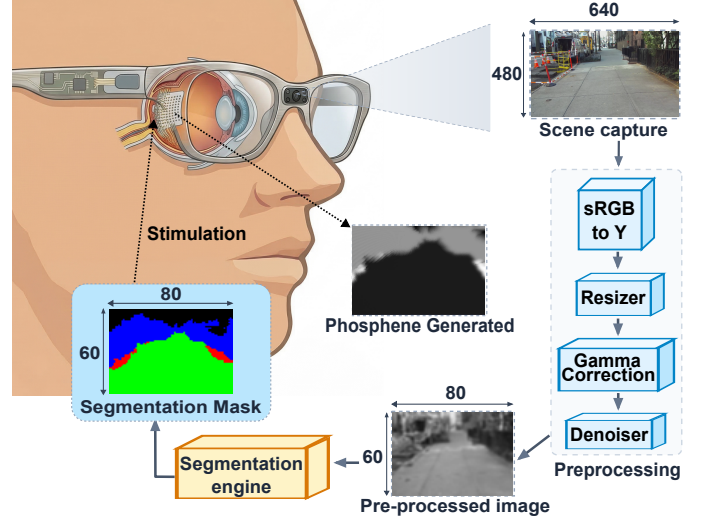


Fig. 1. Proposed spectacle-mounted VPU pipeline. A 640×480 sRGB camera frame is converted to a 60×80 single-channel input, segmented into four navigation classes, and encoded for electrode stimulation.

prosthesis that integrates deep-learning-based navigation assistance.

This paper presents a hardware-aware co-design framework targeting a fully spectacle-mounted VPU with on-chip semantic segmentation. Our contributions are:

- A custom CNN architecture, *NanoSeg*, with 12,880 parameters requiring 12.6 KB of weight storage under INT8 quantisation, for four-class navigation segmentation in retinal prostheses.
- A systematic hardware-aware design methodology demonstrating the trade-offs between segmentation accuracy, model capacity, and on-chip resource consumption across architectural variants.
- FPGA resource utilisation on the Xilinx Zynq-7020 via Vitis HLS C-synthesis, confirming feasibility within wearable-class hardware budgets.

II. RELATED WORK

Beyond the Argus II, other retinal prosthesis approaches include the Alpha AMS subretinal implant [5] and the PRIMA photovoltaic array [6]. These systems relay brightness or contrast information to the electrode array without pixel-level semantic understanding.

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Recent work has explored deep learning as a preprocessing strategy for prosthetic vision. Han et al. [7] showed that semantic segmentation significantly outperforms saliency and depth-based approaches for scene understanding in simulated prosthetic vision, using standard models (DeepLab, DeepGaze II) evaluated in software. Rollend et al. [8] developed CNN-based face detection and object recognition algorithms for the Argus II, executed on an external computer system. Both works validate the utility of deep learning for prosthetic vision but operate on general-purpose hardware.

Lightweight segmentation architectures such as ENet [9] (370 K params), CGNet [10] (0.5 M), and Fast-SCNN [11] (1.1 M) target real-time GPU inference. Hardware-aware co-design efforts, including MCUNet [12] and hls4ml-based FPGA segmentation by Ghielmetti et al. [13] push toward edge deployment, but still requires hundreds of kilobytes of storage.

III. DESIGN CONSTRAINTS

Latency and safety: At pedestrian speeds (1.2–1.5 m/s), a 100 ms delay corresponds to ~ 15 cm of unguided travel. Missed hazard detection directly endangers the user, requiring sub-100 ms inference.

Power and form factor: A spectacle-mounted device must sustain multi-hour operation from a lightweight battery, restricting the design to fixed-point arithmetic (INT8 weights, INT16 accumulators) and on-chip-only memory access.

Memory and input representation: All weights, activations, and buffers must reside in on-chip SRAM. The 640×480 sRGB input is converted to single-channel luminance and downsampled to 60×80 , reducing input size while preserving structural and contrast information for navigation-class discrimination within the on-chip budget.

IV. DATASET AND PREPROCESSING

A. Preprocessing Pipeline

Fig. 1 illustrates the VPU datapath. A 640×480 sRGB camera frame passes through four preprocessing stages before reaching the segmentation model: (i) conversion to a single luminance channel $Y = 0.299R + 0.587G + 0.114B$ (ITU-R BT.601), (ii) spatial downsampling to 60×80 via area averaging, (iii) gamma correction ($\gamma=2.2$) to restore the perceptual luminance distribution expected by the CNN, and (iv) a 3×3 smoothing filter to suppress downsampling artefacts. The segmentation model receives a single-channel 60×80 grayscale image as input.

B. Dataset

Models are trained and evaluated on the SANPO outdoor pedestrian dataset [14], which provides an egocentric video with human-annotated semantic masks—a viewpoint that closely matches a spectacle-mounted camera. We use 40 manually annotated sessions, yielding 3,386 training and 598 test images. Original annotations are mapped to four navigation-relevant classes: *background* (sky, walls), *traversable* (walkable surface), *obstacle* (static/dynamic obstruction), and *hazard* (e.g., road traffic).

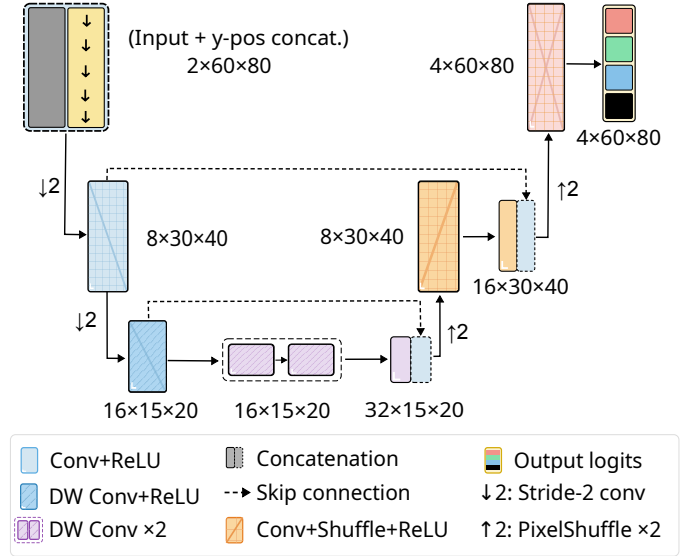


Fig. 2. NanoSeg (V7) architecture — the final design reached after seven iterative refinements. Dual skip connections (enc1 $\rightarrow$ dec2, enc2 $\rightarrow$ dec1) provide U-Net-style multi-scale feature reuse. Block height encodes spatial resolution. All layers require only 3-row line buffers for streaming; vertical position encoding is concatenated at input.

The test set exhibits significant class imbalance: Background 40.8%, Traversable 34.6%, Obstacle 22.2%, Hazard 2.4%. The safety-critical Hazard class is $17\times$ rarer than Background, yet a missed hazard directly endangers the user, motivating aggressive class weighting in the loss function.

V. MODEL ARCHITECTURE AND DESIGN ITERATIONS

The final architecture, NanoSeg (V7), reached after seven iterative refinements described in this section, is shown in Fig. 2.

A. Loss Functions

All variants use weighted cross-entropy

$$\mathcal{L}_{\text{CE}} = - \sum_c w_c y_c \log \hat{p}_c, \quad \mathbf{w} = [0.3, 0.25, 0.7, 3.0], \quad (1)$$

where w_c encodes both class frequency and safety priority. Selected variants augment (1) with a smoothness regulariser penalising abrupt class transitions,

$$\mathcal{L}_{\text{smooth}} = \frac{\lambda}{N} \sum_{i,j} \left[(f_{i,j+1} - f_{i,j})^2 + (f_{i+1,j} - f_{i,j})^2 \right], \quad \lambda=0.1, \quad (2)$$

giving $\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{smooth}}$. One ablation replaces (1) with focal loss [15] with $\gamma=2$, $w_3=4.5$; this variant is labelled V7_F. The loss applied to each variant is stated in the corresponding subsection below.

B. Initial Architecture (V1)

The architecture family, shown in Fig. 2 in its final form, follows a streaming encoder–decoder with PixelShuffle upsampling [16]. The encoder comprises two strided Conv–Batch Normalisation (BN)–ReLU blocks: **enc1** ($60 \times 80 \rightarrow 30 \times 40$, 8

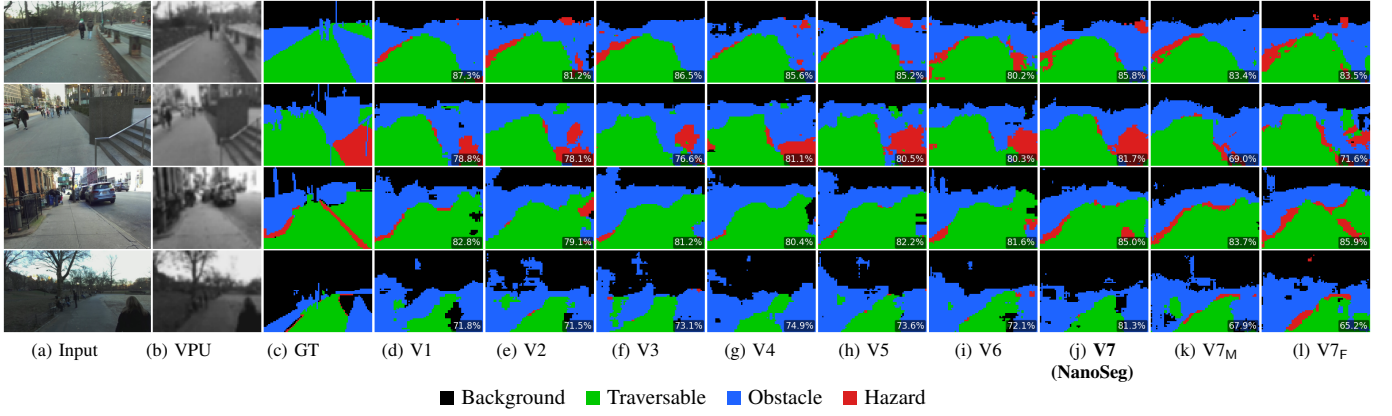


Fig. 3. Qualitative comparison across all model versions on four representative scenes. (a) input 640×480 ; (b) VPU-processed 60×80 ; (c) ground truth; (d)–(j) predictions of V1 through V7 (proposed final); (k) V7_M: 80-session training with manual class weights; (l) V7_F: focal loss ablation. Per-image pixel accuracy shown bottom-right of each prediction tile. V7 achieves the best Hazard IoU while maintaining competitive overall accuracy across four scene types.

channels) and **enc2** ($30 \times 40 \rightarrow 15 \times 20$, 16 channels), followed by a depthwise (DW, groups = channels) bottleneck. The decoder mirrors this with two PixelShuffle upsampling stages: **dec1** ($15 \times 20 \rightarrow 30 \times 40$) and **dec2** ($30 \times 40 \rightarrow 60 \times 80$), each preceded by a $C \rightarrow 4C$ point-wise expansion.

PixelShuffle is chosen over transpose convolution because it requires only a 3-row line buffer, keeping the streaming datapath uniform; full depthwise convolutions reduce the bottleneck MAC count $16 \times$ at the cost of zero inter-channel mixing below enc1. Loss: (1). Parameters: 6,136; mIoU: 0.429.

C. Iterative Architecture Refinements

Seven incremental refinements are summarised in Table I; Fig. 3 shows qualitative outputs across all versions on 4 held-out scenes. For each change we report the *hardware cost* (additional SRAM) and the *accuracy gain* (mIoU Δ), making the design trade-off explicit.

V2 – Shadow Augmentation. Random shadow masks were applied to improve lighting robustness. Loss: (1); no architecture change; 0 B additional SRAM. mIoU fell by 0.039 from V1. The 6,136-parameter model lacked representational capacity to absorb the distribution shift, confirming that capacity must be expanded before augmentation can help.

V3 – Smoothness Loss (+ Shadow Augmentation). Loss: $\mathcal{L}_{CE} + \mathcal{L}_{smooth}$. Smoothness regularisation partially recovered accuracy (Δ mIoU = +0.012 over V2) but did not surpass V1, confirming augmentation as the primary degradation source.

V4 – Smoothness Loss Only. Loss: $\mathcal{L}_{CE} + \mathcal{L}_{smooth}$; no augmentation. mIoU matched V3 (0.430) while Hazard IoU rose to 0.218 (Δ = +0.024 over V3). The smoothness penalty suppresses spatially fragmented predictions—particularly beneficial for the rare Hazard class, where isolated false positives dominate the error term. **Hardware cost: 0 B** (loss-only change).

V5 – Skip Connection and Relaxed Channel Mixing.

Loss unchanged from V4 ($\mathcal{L}_{CE} + \mathcal{L}_{smooth}$). A skip from enc1 to dec2 concatenates multi-scale features into the final upsampling stage; This is streaming-compatible: part of the enc1 map ($30 \times 40 \times 8 = 9,600$ B INT8) is buffered until dec2

reaches the corresponding position. Grouping was relaxed in enc2 (groups = 4) and the bottleneck (groups = 8), restoring inter-channel mixing at modest parameter cost. Δ mIoU = +0.003 over V4. **Hardware cost: +9.4 KB SRAM; +1,440 params.**

V6 – Vertical Position Encoding. Loss unchanged. A precomputed channel $y_{pos} \in [0, 1]$ is concatenated to the grayscale input (+112 params, 0 B extra SRAM), giving the network a static geometric prior: identical luminance at the top of frame predicts Background (sky) while at the bottom it predicts Traversable (ground shadow), resolving a systematic ambiguity in the 60×80 grayscale domain. Δ mIoU = +0.005 over V5. **Hardware cost: 0 B SRAM; +112 params.**

V7 – Dual Skip Connections and Deeper Bottleneck (Final). Loss unchanged. A second skip from enc2 ($15 \times 20 \times 16 = 4,800$ B INT8) to dec1 completes full U-Net-style multi-scale feature reuse. The bottleneck is deepened to two grouped convolutions (groups = 8 then 4), increasing capacity at the most compressed scale where receptive field is largest. Δ mIoU = +0.019 over V6; all four per-class IoUs improve. **Hardware cost: +4.7 KB SRAM; +4,968 params.** NanoSeg is Pareto-optimal: no other version achieves both >0.45 mIoU and >0.20 Hazard IoU (Table I).

Extended ablations. Training V7 on 80 sessions with manual class weights raised raw accuracy marginally (68.8%) but collapsed Hazard IoU to 0.087: the enlarged dataset diluted hazard-region supervision below the threshold where weighted-CE can compensate. Substituting focal loss (??) ($\gamma=2$, $w_3=4.5$) similarly failed (Hazard IoU 0.078) — down-weighting easy negatives conflicts with extreme class imbalance at 60×80 , where most Hazard pixels neighbour easy-to-classify Traversable regions. Both results reinforce the 40-session weighted-CE configuration.

VI. TRAINING SETUP AND RESULTS

A. Training Setup

All models are implemented in PyTorch. The Adam optimiser [17] is used with an initial learning rate of 10^{-3} , decayed

via cosine annealing to 10^{-5} over $T_{\max} = 300$ epochs. Mini-batches of 32 images are drawn from 3,386 training frames. Each image passes through the VPU pipeline (Section IV-A) to produce a 60×80 single-channel grayscale input before ingestion by the model; no test-time augmentation is applied. The 598-image test split (seed=42) is held out throughout; the best checkpoint is selected by test mIoU.

B. Evaluation Metrics

Let TP_c , FP_c , FN_c denote true positives, false positives, and false negatives for class c . Per-class IoU and its mean (mIoU) are the primary accuracy metrics:

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad mIoU = \frac{1}{C} \sum_{c=1}^C IoU_c. \quad (3)$$

One safety-critical metric are additionally reported. **Hazard Recall (Hazz-R)** measures the fraction of ground-truth Hazard pixels correctly detected:

$$R_{Hz} = \frac{TP_{Hz}}{TP_{Hz} + FN_{Hz}}. \quad (4)$$

C. Iterative Model Results

Table I reports all metrics across the design trajectory. Excluding the unsuccessful augmentation experiments (V2–V3), mIoU improves monotonically from 0.429 (V1) to 0.464 (V7), while Hazard Recall rises from 0.462 to 0.530 — a +14.7% relative gain on the safety-critical class. The 80-session ablations collapse Hazard Recall to 0.106 and 0.233 respectively, validating the 40-session weighted-CE configuration. Fig. 3 shows qualitative outputs across four representative scenes.

D. Qualitative Results

Fig. 4 presents NanoSeg inference across six scenes at three difficulty levels. Easy scenes yield clean traversable/background separation. Medium scenes with mixed obstacles and pedestrians are correctly navigated. Hard scenes — low contrast, shadows, multiple simultaneous hazards — represent the most demanding conditions for a prosthesis user; the model correctly segments the traversable corridor in all cases. The overlay column shows the signal delivered to the electrode array after segmentation encoding.

E. Comparison with State-of-the-Art

Table II compares NanoSeg against four models fine-tuned on identical VPU-processed 60×80 input. General-purpose models achieve higher mIoU (0.631–0.742 vs. 0.464) owing to their $57 \times$ – $1,700 \times$ parameter advantage, but none fits within the 29 KB ASIC budget: even ENet exceeds it by $100 \times$ and would require >100 mW at 65 nm — incompatible with wearable constraints. NanoSeg achieves 35.97 mIoU per million parameters, $21 \times$ higher than ENet (1.71), demonstrating that architecture co-design under ASIC constraints yields efficiency gains far beyond parameter reduction alone.

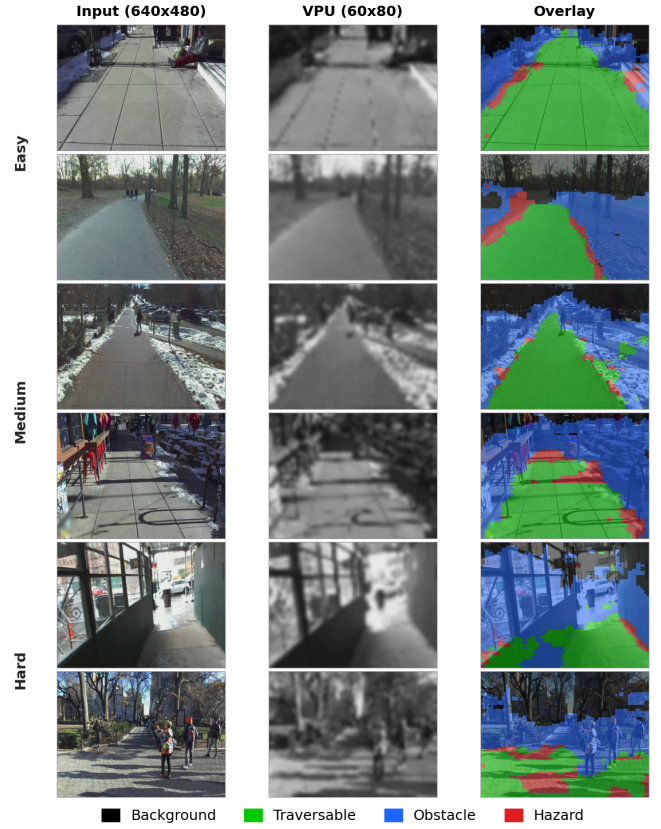


Fig. 4. NanoSeg inference on six outdoor scenes (rows 1–2: easy; 3–4: medium; 5–6: hard). Columns: input 640×480 ; VPU-processed 60×80 (upscaled for display); segmentation overlay (50% blend).

VII. HARDWARE-AWARE ANALYSIS

A. Post-Training Quantisation

We quantize all weights to INT8 (symmetric per-channel) with INT16 accumulators. Nine per-layer scale factors are stored as 16-bit fixed-point. The quantised weight footprint is 12,856 B (12.6 KB) — a $4 \times$ reduction over FP32 with negligible accuracy degradation.

B. On-Chip Memory Analysis

Streaming eliminates full-frame buffers; each layer retains at most three rows of its input feature map. The dominant buffer is dec1 ($3 \times 20 \times 32 = 1,920$ B); enc2 and enc1 skip buffers add 960 B each. Table III gives the complete breakdown; the total is **29.0 KB**, feasible within sub-mW ASIC on-chip SRAM.

C. FPGA Synthesis Methodology

NanoSeg was compiled with Vitis HLS 2020.2 [21] targeting the Xilinx Zynq-7020 at 100 MHz. Three adaptations were required: PixelShuffle replaced with nearest-neighbour upsampling + Conv2d; position encoding baked as a compile-time constant; and BatchNorm folded into convolutional biases. Kernel loops were pipelined at $\Pi=1$; feature maps mapped to LUT-RAM (BIND_STORAGE impl=lutram) with channel-dimension array partitioning for parallel access.

TABLE I

ITERATIVE MODEL COMPARISON ON SANPO TEST SET (40 SESSIONS). PER-CLASS COLUMNS REPORT IOU; HAZD-R: HAZARD RECALL — FRACTION OF GROUND-TRUTH HAZARD PIXELS CORRECTLY DETECTED; THE PRIMARY SAFETY METRIC.

Ver.	Architecture Change	mIoU	Bkgd	Trav	Obst	Hazd	Hazd-R	Params
V1	Initial: full-DW enc-dec, PixelShuffle	0.429	0.532	0.609	0.382	0.194	0.462	6,136
V2	+ Shadow augmentation	0.390	0.485	0.561	0.359	0.154	0.464	6,136
V3	+ Smoothness loss + shadow augmentation	0.402	0.492	0.578	0.367	0.173	0.428	6,136
V4	+ Smoothness loss only	0.430	0.521	0.606	0.377	0.218	0.466	6,136
V5	+ Skip enc1→dec2, relaxed DW grouping	0.433	0.535	0.614	0.377	0.208	0.474	7,576
V6	+ Vertical position encoding	0.438	0.549	0.619	0.382	0.201	0.455	7,648
V7 (NanoSeg)	+ Dual skip (enc2→dec1), deeper bottleneck	0.464	0.584	0.630	0.400	0.213	0.530	12,616
V7-80s	V7 + 80-session training, manual weights	0.418	0.601	0.627	0.358	0.087	0.106	12,616
V7-focal	V7 + 80-session training, focal loss	0.403	0.586	0.599	0.348	0.078	0.233	12,616

TABLE II

COMPARISON WITH SOTA SEGMENTATION MODELS ON THE SANPO VALIDATION SET (60×80 INPUT, 598 IMAGES).

Model	Params	mIoU	mIoU/M [†]	Mem	ASIC
SegFormer-B0 [18]	3.72M	0.728	0.20	>14 MB	✗
ERFNet [19]	2.06M	0.742	0.36	>8 MB	✗
DeepLabV3+ [20]	11.0M	0.672	0.06	>42 MB	✗
ENet [9]	0.37M	0.631	1.71	>3 MB	✗
NanoSeg V7 (Ours)	12.9K	0.464	35.97	29 KB	✓

[†]mIoU per million params. Baselines pretrained on ADE20K/Cityscapes/ImageNet, finetuned on same split. Baseline memory exceeds ASIC budget by 100–1400×, requiring off-chip DRAM infeasible for wearable deployment.

TABLE III

ON-CHIP MEMORY BREAKDOWN (INT8, STREAMING DATAPATH)

Component	Size
Quantised weights (12,856 params × 1 B)	12.6 KB
Scale factors (9 layers × 2 B)	0.02 KB
Active streaming line buffers	15.4 KB
Skip-connection buffers (enc1 + enc2)	1.0 KB
Total	29.0 KB

D. Synthesis Results

Table IV summarises C-synthesis estimates. The critical path is 7.28 ns, meeting the 10 ns target with 2.72 ns margin. A full 60×80 inference takes 6,720,068 cycles (67.2 ms at 100 MHz, ≈14.9 fps), within the ~100 ms prosthesis constraint.

LUT usage (66%) is dominated by dec1 (32 channels, 9,216 params), while only 6 DSP48s are used as smaller ops map to LUTs. BRAM (46%) stores weights, and low FF usage (6%) indicates a memory-bound design. Post place-and-route may vary by 10–20%.

VIII. CONCLUSION

This paper presented NanoSeg, a 12,880-parameter CNN for four-class navigation segmentation in epiretinal prostheses, designed through seven iterative architectural refinements

TABLE IV

VITIS HLS C-SYNTHESIS RESOURCE ESTIMATES (ZYNQ-7020, 100 MHz)

Resource	Used	Available	Util.
BRAM_18K	130	280	46%
DSP48E	6	220	2%
Flip-Flop	6,843	106,400	6%
LUT	35,469	53,200	66%
Critical path			7.28 ns
Inference latency			67.2 ms

under ASIC memory and compute constraints. The model achieves mIoU 0.464 with 12.6 KB INT8 weights, and FPGA C-synthesis on the Zynq-7020 confirms 67.2 ms inference at 100 MHz, validating deployability within wearable hardware budgets. To our knowledge, this is the first work to demonstrate semantic segmentation at the sub-30 KB memory scale required for prosthesis-integrated deployment. Future work targets a custom streaming ASIC in 65 nm CMOS and closed-loop integration with the epiretinal electrode driver.

REFERENCES

- [1] W. L. Wong, X. Su, X. Li, C. M. G. Cheung, R. Klein, C.-Y. Cheng, and T. Y. Wong, “Global prevalence of age-related macular degeneration and disease burden projection for 2020 and 2040: a systematic review and meta-analysis,” *The Lancet Global Health*, vol. 2, no. 2, pp. e106–e116, 2014.
- [2] D. T. Hartong, E. L. Berson, and T. P. Dryja, “Retinitis pigmentosa,” *The Lancet*, vol. 368, no. 9549, pp. 1795–1809, 2006.
- [3] L. da Cruz, J. D. Dorn, M. S. Humayun, G. Dagnelie, J. Handa, P.-O. Barale, J.-A. Sahel, P. E. Stanga, F. Hafezi, A. B. Safran, *et al.*, “Five-year safety and performance results from the Argus II retinal prosthesis system clinical trial,” *Ophthalmology*, vol. 123, no. 10, pp. 2248–2254, 2016.
- [4] Second Sight Medical Products, “Second sight announces plans to wind down operations,” Press release, 2020. The company later merged with Nano Precision Medical in 2022, but Argus II production and support were not resumed.
- [5] K. Stingl *et al.*, “Interim results of a multicenter trial with the new electronic subretinal implant Alpha AMS in 15 patients blind from inherited retinal degenerations,” *Frontiers in Neuroscience*, vol. 11, p. 445, 2017.
- [6] D. Palanker *et al.*, “Photovoltaic restoration of central vision in atrophic age-related macular degeneration,” *Ophthalmology*, vol. 127, no. 8, pp. 1097–1104, 2020.

- [7] N. Han, S. Srivastava, A. Xu, D. Klein, and M. Beyeler, "Deep learning-based scene simplification for bionic vision," in *Proc. ACM Augmented Humans (AHs)*, pp. 45–54, 2021.
- [8] D. Rollend, P. Rosendall, S. Billings, P. Burlina, K. Wolfe, and K. Katyal, "Face detection and object recognition for a retinal prosthesis," in *ACCV 2016 Workshops, LNCS*, vol. 10116, pp. 303–313, Springer, 2017.
- [9] A. Paszke, A. Chaurasia, S. Kim, and E. Culurciello, "ENet: A deep neural network architecture for real-time semantic segmentation," *arXiv preprint arXiv:1606.02147*, 2016.
- [10] T. Wu, S. Tang, R. Zhang, J. Cao, and Y. Zhang, "CGNet: A light-weight context guided network for semantic segmentation," *IEEE Transactions on Image Processing*, vol. 30, pp. 1169–1179, 2021.
- [11] R. P. K. Poudel, S. Liwicki, and R. Cipolla, "Fast-SCNN: Fast semantic segmentation network," *arXiv preprint arXiv:1902.04502*, 2019.
- [12] J. Lin, W.-M. Chen, Y. Lin, C. Gan, and S. Han, "MCUNet: Tiny deep learning on IoT devices," in *Proc. NeurIPS*, 2020.
- [13] N. Ghielmetti *et al.*, "Real-time semantic segmentation on FPGAs for autonomous vehicles with hls4ml," *Machine Learning: Science and Technology*, vol. 3, no. 4, p. 045011, 2022.
- [14] V. Blanchard *et al.*, "Sanpo: A scene understanding, accessibility, navigation, pathfinding, obstacle dataset," *arXiv preprint arXiv:2309.03254*, 2023.
- [15] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE ICCV*, pp. 2980–2988, 2017.
- [16] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE CVPR*, pp. 1874–1883, 2016.
- [17] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [18] E. Xie *et al.*, "Segformer: Simple and efficient design for semantic segmentation with transformers," in *Proc. NeurIPS*, 2021.
- [19] E. Romera *et al.*, "Erfnet: Efficient residual factorized convnet for real-time semantic segmentation," *IEEE Trans. Intelligent Transportation Systems*, 2017.
- [20] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. ECCV*, pp. 801–818, 2018.
- [21] Xilinx Inc., *Vitis High-Level Synthesis User Guide (UG1399)*, 2020.